Jason B. Hill

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Technical Stack

Languages & Frameworks: Python (Jupyter, Numpy, Pandas, Dash, SciPy), Clojure, Bash, R, Java (Micronaut, Spring)

Data & Infrastructure: PostgreSQL (window functions, UDFs, PL/pgSQL, recursive CTEs), BigQuery, GCP (Google Logging, Pub/Sub, GCS), Kafka, Splunk, Spark SQL, Docker, architecture (Kimball/star, snowflake, xNF)

Visualization Frameworks: Looker, Google Logging, Splunk, Dash, Plotly, R-Shiny, Seaborn, Matplotlib

AI/ML: Vertex AI (Gemini), OpenAI, Cursor (Claude Sonnet/Opus, Gemini Code Assist), Scikit-learn, TensorFlow

Professional Experience

Rally Software — Data Engineer, Senior Data Engineer
CA Agile Central — Data Scientist, Senior Data Scientist, Principal Data Scientist
Broadcom (Rally – Agile Operations Division) — Staff Data Scientist
Broomfield, Denver, Boulder, CO — Oct 2015 – Oct 2025

- Led technical side of development for Agile flow metrics and Agile forecasting products.
- Developed a "Smart" Cumulative Flow Diagram for analyzing team and train delivery metrics.
- Led team and train-level portfolio flow metrics design and development.
- Used Vertex AI (Gemini) to produce flow metrics chart interpretations and coaching advice for teams.
- Spearheaded capacity and delivery forecasting development for team and train-level planning.
- Helped develop, maintain, and use a distributed tracing infrastructure for a visible and reliable usage and performance metrics system to report 100% of user interactions in a SaaS environment involving 250K+ monthly users. Used the resulting data with Splunk, BigQuery, Looker, Spark, and other systems to analyze interactions.
- Led performance optimization, via tracing and visualizations, for Rally's Oracle to PostgreSQL migration.
- Built and maintained a "product data hub" for those within Rally to track feature adoption, subscription usage, performance, and user input.
- Built a leading indicator for subscription renewal predictions using usage data and random forest models.
- Worked in Python, Java, Clojure, Bash, Scala, JavaScript (React), and elsewhere as needed.

SpotXChange — Data Scientist
Broomfield, CO — Apr 2014 – Sep 2015

- Developed a large-scale Hive datastore for 600+ frontend ad-tech bidding servers using Python and Bash.
- Tracked automated view bidding for website advertisements.
- Led development of fraud activity investigations using big data, Hadoop, ETL, MySQL, and Tableau.
- Built Tableau dashboards for product and C-level insights.

Silicon Graphics, Inc. (SGI) — Software Engineer
Longmont, CO — Jun 2013 – Mar 2014

- Led development and testing of SCSI Enclosure Service (SES) Python/Cython module.
- Worked on various firmware/control modules in C, C++ (Boost), Python, and Cython.

Education

Ph.D. in Mathematics (2014/ABD) University of Colorado (Alexander Hulpke and Keith Kearnes)

- Computational group and graph theory, permutation group algorithms, NCAR & XSEDE supercomputers
- Developed a massively parallel approach to solving partition backtrack problems in group and graph theory.

M.A. in Mathematics – 2010 University of Colorado (Keith Kearnes)

- Computational group theory, representation theory

M.S. in Mathematics – 2006 University of Vermont (David S. Dummit)

- Computational algebraic number theory
- Developed an algorithm for finding number fields satisfying given properties and having mimimum discriminant. Found the first known instance of a totally real quintic number field having minimal signature group rank.

B.S. in Mathematics – 2004 (logic, minor in physics) University of Michigan–Flint

- I studied abroad in Uppsala, Sweden for one year. Instead of taking usual foreign student classes, I took math and philosophy courses in Swedish. I knew a bit of Swedish before (not much). I passed course exams.
- President of student groups: Mathematics Club, Students for Social Change

Miscellaneous

Volunteering

- In my spare time (in 2019, 2020, 2024, and 2025), I volunteer with the Save Elephant Foundation and Elephant Nature Park in Northern Thailand to help rescue and care for abused elephants.

Meetups and Knowledge Sharing

- I attend and have spoken four times at local Agile Meetup groups (Agile Boulder, Agile Denver, Agile Denver North), discussing: Cumulative Flow Diagrams (CFDs), probability and "wisdom-of-the-crowds" versus "group-think," Monte Carlo simulations for Agile forecasting, and some details about Little's Law.
- I coordinated the Denver/Boulder Type-1 Diabetic Adults Meetup from early 2017 until early 2020.
- I helped coordinate and spoke three times at the Analyze Boulder Meetup.
- Splunk invited me to give a talk in 2018. I presented on distributed tracing, ingestion into Splunk, and the resulting data analysis and visualizations that can be carried out on the platform.
- I coordinated the Denver Area Twitter Developer Community Meetup from October of 2016 until early 2020. I was not employed by Twitter, but we worked together to spread info about their APIs and analysis of Twitter data.
- Since resigning from Rally, my goal is to share some Agile metrics knowledge, specifically related to Little's Law and forecasting (things beyond the scope of my work at Rally), via my website: www.jasonbhill.com. My current goal is to have this online at some point in March of 2026.

Other Employment

- In addition to teaching many courses in mathemtics while attending graduate school, I've held two adjunct instructor positions, one at Saint Michael's College in mathematics and one at The Community College of Vermont in photography.
- I've worked multiple times as a contract photographer for Automobile Magazine.